

Government Polytechnic Gandhinagar

Information Technology Department

Assignment – Unit 1

Course: Operating Systems (DI03016061)

Semester: 3

Total Marks: 30

Submission: Handwritten with diagrams wherever required.

Part A – Conceptual Questions (10 Marks)

Q1. Define an Operating System. Explain any **five functions** performed by an Operating System.

Q2. Explain the **evolution of Operating Systems** with suitable examples.

Q3. Differentiate between:

1. Batch OS and Time-Sharing OS
 2. Multiprogramming and Multiprocessing OS
 3. Multitasking and Real-Time OS
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Part B – Logical Thinking Questions (10 Marks)

Q4. Case Study

A college computer laboratory has the following requirements:

- 80 students work simultaneously.
- Many applications should run together.
- Students should receive quick responses.
- The lab is connected to a central server.

Answer the following:

- a) Which Operating System type is most suitable? Why?
- b) Which OS service helps students open files and save their work?
- c) Which component (Kernel/Shell/System Call) communicates directly with hardware?

d) Which OS service manages printers and USB devices?

Q5. Match the Situation with the Correct OS Type

Situation	Operating System Type
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Railway Reservation System	_____
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ATM Machine	_____
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Android Mobile	_____
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College Server	_____
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Payroll Processing	_____
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Give a one-line reason for each answer.

Part C – Diagram and Analysis (5 Marks)

Q6.

Draw and explain the architecture of an Operating System showing:

- User
- Application Program
- Shell
- Kernel
- Hardware

Also explain the role of each component.

Part D – Application-Based Questions (5 Marks)

Q7.

A user types:

mkdir OS

Explain step-by-step what happens inside the Operating System until the directory is created.

Include:

- User
 - Shell
 - System Call
 - Kernel
 - File System
 - Hardware
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Q8.

Fill the following table.

Activity	Kernel Shell System Call
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Accepts user command	
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Allocates memory	
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Creates file	
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Communicates with hardware	
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Provides command interface	
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